

Practical 1 and 2

1)

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mahij@Mahija: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    char cmd[100];  
  
    printf("Enter Linux command: ");  
    scanf("%s", cmd);  
  
    pid_t pid = fork();  
  
    if (pid == 0)  
    {  
        printf("Child PID : %d\n", getpid());  
        execlp(cmd, cmd, NULL);  
        perror("Execution Failed");  
    }  
    else if (pid > 0)  
    {  
        printf("Parent PID : %d\n", getpid());  
        wait(NULL);  
        printf("Child process completed.\n");  
    }  
    else  
    {  
        printf("Fork Failed\n");  
    }  
  
    return 0;  
}
```

```

mahij@Mahija:~$ nano linux.c
mahij@Mahija:~$ gcc linux.c -o linux
mahij@Mahija:~$ ./linux
Enter Linux command: uname
Parent PID : 496
Child PID : 569
Linux
Child process completed.
mahij@Mahija:~$ |

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```

mahij@Mahija:~$ gcc linux.c -o linux
mahij@Mahija:~$ ./linux
Enter Linux command: lscpu
Parent PID : 622
Child PID : 625
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      12
On-line CPU(s) list:         0-11
Vendor ID:                   GenuineIntel
Model name:                  13th Gen Intel(R) Core(TM) i5-1334U
CPU family:                   6
Model:                       186
Thread(s) per core:          2
Core(s) per socket:          6
Socket(s):                    1
Stepping:                     3
BogoMIPS:                     4992.02
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
cp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bml avx2 smep bmi2 erms invpcid rdseed adx smap clflush
opt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
clear serialize ibt flush_l1d arch_capabilities

Virtualization features:
Virtualization:              VT-x
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                          288 KiB (6 instances)
L1i:                          192 KiB (6 instances)
L2:                           7.5 MiB (6 instances)
L3:                           12 MiB (1 instance)
NUMA:
NUMA node(s):                 1
NUMA node0 CPU(s):            0-11
Vulnerabilities:
Gather data sampling:         Not affected
Ghostwrite:                   Not affected
Indirect target selection:    Not affected
Itlb multihit:                Not affected
L1tf:                         Not affected
Mds:                          Not affected
Meltdown:                     Not affected
Mmio stale data:              Not affected
Old microcode:                Not affected
Reg file data sampling:       Mitigation; Clear Register File
Retbleed:                     Mitigation; Enhanced IBRS
Spec rstack overflow:         Not affected
Spec store bypass:            Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:                   Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:                   Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds:                         Not affected
Tsx:                          Not affected
Tsx async abort:              Not affected
Vmscape:                      Not affected
Child process completed.
mahij@Mahija:~$ |

```

```

mahij@Mahija:~$ gcc linux.c -o linux
mahij@Mahija:~$ ./linux
Enter Linux command: lsblk
Parent PID : 653
Child PID : 656
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0      0 356.9M  1 disk
sdb   8:16     0 159.4M  1 disk
sdc   8:32     0    2G    0 disk [SWAP]
sdd   8:48     0    1T    0 disk /mnt/wslg/distro
/

Child process completed.
mahij@Mahija:~$ |

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mahij@Mahija:~$ gcc linux.c -o linux
mahij@Mahija:~$ ./linux
Enter Linux command: ps
Parent PID : 680
Child PID : 681
  PID TTY          TIME CMD
  310 pts/0        00:00:00 bash
  680 pts/0        00:00:00 linux
  681 pts/0        00:00:00 ps
Child process completed.
mahij@Mahija:~$ |

```

```

mahij@Mahija:~$ gcc linux.c -o linux
mahij@Mahija:~$ ./linux
Enter Linux command: top
Parent PID : 710
Child PID : 711
top - 16:28:46 up 15 min,  1 user,  load average: 0.03, 0.03, 0.00
Tasks:  25 total,   1 running,  24 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem :  7802.6 total,  7180.7 free,   555.3 used,   217.7 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7247.3 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24128	15396	11604	S	0.0	0.2	0:01.10	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
6	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
49	root	19	-1	50428	16724	15448	S	0.0	0.2	0:00.27	systemd-journal
81	systemd+	20	0	22540	14576	11960	S	0.0	0.2	0:00.12	systemd-resolve
88	root	20	0	35236	12328	9252	S	0.0	0.2	0:00.72	systemd-udev
110	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.06	chronyd-starter
111	root	20	0	4472	3140	2892	S	0.0	0.0	0:00.01	cron
112	message+	20	0	8820	5416	4740	S	0.0	0.1	0:00.08	dbus-daemon
156	root	20	0	44092	29732	14592	S	0.0	0.4	0:00.38	networkd-dispat
160	root	20	0	18436	9408	8160	S	0.0	0.1	0:00.18	systemd-logind
178	syslog	20	0	220548	5852	4592	S	0.0	0.1	0:00.09	rsyslogd
224	_chrony	20	0	20420	6444	5668	S	0.0	0.1	0:00.07	chronyd
226	_chrony	20	0	12092	2400	1760	S	0.0	0.0	0:00.00	chronyd
260	root	20	0	123456	32624	18012	S	0.0	0.4	0:00.24	unattended-upgr
275	root	20	0	5492	2732	2548	S	0.0	0.0	0:00.02	agetty
307	root	20	0	3184	1108	980	S	0.0	0.0	0:00.00	SessionLeader
308	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.05	Relay(310)
310	mahij	20	0	6268	5532	3856	S	0.0	0.1	0:00.12	bash
311	root	20	0	8644	5540	4700	S	0.0	0.1	0:00.00	login
375	mahij	20	0	22332	13588	10968	S	0.0	0.2	0:00.17	systemd
377	mahij	20	0	22912	4128	2124	S	0.0	0.1	0:00.00	(sd-pam)
406	mahij	20	0	6136	5480	3928	S	0.0	0.1	0:00.03	bash
710	mahij	20	0	2768	1876	1780	S	0.0	0.0	0:00.00	linux
711	mahij	20	0	8192	6064	3940	R	0.0	0.1	0:00.01	top

2)

```
mahij@Mahija: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <fcntl.h>  
#include <unistd.h>  
  
int main()  
{  
    int src, dest;  
    char buffer[1024];  
    int bytes;  
  
    src = open("input.txt", O_RDONLY);  
  
    if(src < 0)  
    {  
        printf("Cannot open source file\n");  
        return 1;  
    }  
  
    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);  
  
    while((bytes = read(src, buffer, sizeof(buffer))) > 0)  
    {  
        write(dest, buffer, bytes);  
    }  
  
    close(src);  
    close(dest);  
  
    printf("File copied successfully.\n");  
  
    return 0;  
}
```



mahij@Mahija: ~



```
mahij@Mahija:~$ nano copy.c
mahij@Mahija:~$ gcc copy.c -o copy
mahij@Mahija:~$ ./copy
Cannot open source file
mahij@Mahija:~$ echo "Hello, Ubuntu!" > input.txt
mahij@Mahija:~$ ./copy
File copied successfully.
mahij@Mahija:~$ |
```